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# simple_viz – project handoff
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A complete, self-contained guide to this assignment so you can continue any time, even without the original chat. Read top to bottom once; after that use it as a reference.

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1. What this project is

simple_viz is a small Python **data-visualization library** built for a communication course. It makes "communication-first" charts that tell one data story – **Pinterest's 2025 results** – where every chart is the right chart for one specific question and makes deliberate design choices.

The assignment has two graded parts:

1. **The library** – a real, installable, importable Python package.
2. **A ~2-minute video** demoing it (see §8 and `VIDEO_SCRIPT.md`).

The core idea / thesis of the data story: Pinterest has a huge, fast-growing *global* audience, but its *revenue* is heavily concentrated in the US & Canada – "users are global, revenue is domestic."

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2. Folder structure

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Comms_Data_Science/
├─ src/simple_viz/          # THE LIBRARY (this is what's graded)
│   ├── __init__.py        # public API: exposes the 5 chart functions + __version__
│   ├── core.py            # the 5 chart functions (the real logic)
│   └─ theme.py            # shared colours + styling helpers (one design system)
├─ tests/
│   └─ test_simple_viz.py  # 9 pytest tests (return types + 2 calculations)
├─ data/                   # real, sourced Pinterest data (CSV)
│   ├── pinterest_mau.csv  # annual Q4 MAU 2019–2025
│   ├── pinterest_mau_quarterly.csv # quarterly MAU (28 points) 2019–2025
│   ├── pinterest_regions_q4_2025.csv # MAU + revenue by region (Q4 2025)
│   ├── pinterest_revenue.csv # annual revenue 2019–2025
│   └─ SOURCES.md          # where every number comes from (links)
├─ examples/               # DEMOS (not part of the library)
│   ├── simple_viz_gallery.py # renders the 5 charts + a combined PDF
│   ├── simple_viz_poster.py  # the one-page infographic poster
│   ├── simple_viz_animation.py # animated growth line → simple_viz_growth.gif
│   ├── simple_viz_story_animation.py # full story animated → simple_viz_story.gif
│   ├── simple_viz_01_big_number.png ... 05_revenue_per_user.png # the 5 chart images
│   ├── simple_viz_gallery.pdf # 5 charts combined
│   ├── simple_viz_poster.png / .pdf # the poster
│   ├── simple_viz_growth.gif # animated growth line
│   └─ simple_viz_story.gif # animated full story
└─ README.md               # install + usage + the function/chart-type table
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└─ ANALYSIS.md                # the business "so what": purpose, insights, decisions
└─ VIDEO_SCRIPT.md            # timed 2-minute recording script
└─ HANDOFF.md                 # this file
└─ pyproject.toml             # package metadata + dependencies
└─ LICENSE                     # MIT
└─ .github/workflows/python-publish.yml # (auto-added) publish-to-PyPI on release
...`
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****Install name vs import name:**** the package installs as `simple\_viz` and imports as `simple\_viz`. Version is `0.1.0` (first working version).

3. The library API (the 5 chart functions)

All live in `src/simple\_viz/core.py`. Each takes a pandas DataFrame (or simple values) and ****returns a matplotlib `Figure`**** you can `savefig` in any format.

Function	Chart type	What it answers	Key call
`big_number(value, unit, label, footnote="")`	Single important number	How big is the headline?	`simple_viz.big_number(619, "M", "monthly active users")`
`growth_line(df, x, y, title, annotate=None)`	Change over time	Which way is the trend going?	`simple_viz.growth_line(mau, "year", "maus_millions", title=..., annotate=(2021, "dip"))`
`revenue_bar(df, category, value, title, spotlight=None)`	Comparison	Who contributes the most?	`simple_viz.revenue_bar(reg, "region", "revenue_musd", title=..., spotlight="US & Canada")`
`share_gap(df, category, part_a, part_b, title, highlight=None)`	Parts of a whole	Users vs revenue mismatch	`simple_viz.share_gap(reg, "region", "maus_millions", "revenue_musd", title=..., highlight="Rest of World")`
`revenue_per_user(df, category, revenue, users, title, spotlight=None)`	Comparison / ratio	How well is each region monetised?	`simple_viz.revenue_per_user(reg, "region", "revenue_musd", "maus_millions", title=...)`

4. How the code works (module by module)

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### `src/simple_viz/theme.py` – one shared design system
Holds the palette (`PIN_RED = #E60023`, plus muted greys `MUTED`, `INK`, `SUBTLE`, `GRID`) and three helpers every chart reuses:
- `apply_base_style(ax)` – removes chartjunk (drops top/right spines, softens the rest, kills tick marks, lightens the y-gridlines).
- `titles(ax, title, subtitle)` – left-aligned bold **takeaway** title + a lighter subtitle (leads with the message, not the variable name).
- `new_axes(figsize)` – makes a figure/axes with headroom for the titles.
- It also sets `plt.rcParams["text.parse_math"] = False` – see §7, problem 2.
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### `src/simple_viz/core.py` – the 5 functions
Each function: builds a figure, draws its chart, applies the shared theme, and returns the `Figure`. `revenue_bar` and `revenue_per_user` share one private helper
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`\_spotlight\_barh` (sorted bars, one highlight colour, end-of-bar labels) so there's no duplicated code. `revenue\_per\_user` computes revenue ÷ users itself so the ratio is never hand-entered.

`src/simple\_viz/\_\_init\_\_.py` – the public face
Imports the 5 functions so users can call `simple\_viz.growth\_line(...)` and defines `\_\_all\_\_` and `\_\_version\_\_`.

The design rules every chart follows

1. **One highlight colour** (Pinterest red); everything else muted grey – colour directs attention, doesn't decorate.
2. **Takeaway titles** – "Rest of World is 58% of users but 7% of revenue," not "Users and revenue by region."
3. **Honest axes** – bars/lines start at zero; shares normalised to 100%.
4. **No chartjunk** – no top/right spines, no tick marks, light gridlines, values labelled directly (no legends).

5. The visualizations produced

Five library charts (run `python examples/simple\_viz\_gallery.py`):

1. `simple\_viz\_01\_big\_number.png` – **619M** big-number callout.
2. `simple\_viz\_02\_growth\_line.png` – MAU growth 2019–2025, 2021 dip annotated.
3. `simple\_viz\_03\_revenue\_bar.png` – Q4 revenue by region (US & Canada spotlighted).
4. `simple\_viz\_04\_share\_gap.png` – users-vs-revenue split (Rest of World highlighted).
5. `simple\_viz\_05\_revenue\_per\_user.png` – implied revenue per user by region.

One composed poster (`python examples/simple\_viz\_poster.py`):

- `simple\_viz\_poster.png` / `.pdf` – a one-page infographic: red kicker → serif headline → 619M / \$4.2B KPIs → quarterly growth line → revenue-growth bars → the user/revenue split → revenue by region → revenue per user → a red insight band ("a US & Canada user is worth about 35× a Rest-of-World user").
Beige background, white cards, one accent colour.

Two animations (optional creative extras):

- `simple\_viz\_growth.gif` – the growth line drawing in (`simple\_viz\_animation.py`).
- `simple\_viz\_story.gif` – the full story building in sequence
(`simple\_viz\_story\_animation.py`).

6. Data & sources

All numbers are **real**, from Pinterest's Q4 & Full-Year 2025 report (released Feb 12, 2026) and its 10-K. They were transcribed into the CSVs in `data/` and loaded with `pandas.read\_csv` – there is **no live API call**. Full source links are in **`data/SOURCES.md`**. Key figures:

- 619M global MAU (+12% YoY); \$4.22B FY revenue (+16%); \$1.319B Q4 revenue.
- By region (Q4 2025): US & Canada 105M MAU / \$979M; Europe 158M / \$245M; Rest of World 356M / \$96M.
- Annual revenue 2019→2025: \$1.14B → \$4.22B; global ARPU \$3.81 → \$7.21.

See `**`ANALYSIS.md`**` for what the charts **mean** – the business insights and decisions they support (invest in monetising Rest of World / Europe rather than chasing more saturated US users; manage to ARPU-by-region, not just MAU).

7. Two problems solved (for the video's "problems" section)

1. `**`ModuleNotFoundError`` on ``import simple_viz``. Cause: the modern ``src/`` layout means Python can't find the package until it's installed. **\*\*Fix:\*\*** ``python -m pip install -e .`` from the project root (editable install links the package so imports resolve and edits apply live).
2. ****Dollar-sign labels rendered garbled**** (e.g. ``$9 ... $0.27``). Cause: `matplotlib` treats a pair of ``$`` as LaTeX math mode. ****Fix:**** ``plt.rcParams["text.parse_math"] = False`` (set once in ``theme.py``).

8. How to run everything (commands)

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from the project root (folder with pyproject.toml)

python -m pip install -e . # install the library (editable)
python -c "import simple_viz; print(simple_viz.__all__)" # confirm it imports

python -m pip install -e ".[dev]" # install pytest too
python -m pytest -q # run the 9 tests (should pass)

python examples/simple_viz_gallery.py # render the 5 charts + gallery PDF
python examples/simple_viz_poster.py # render the poster (PNG + PDF)
python examples/simple_viz_animation.py # render simple_viz_growth.gif
python examples/simple_viz_story_animation.py # render simple_viz_story.gif

build a distributable package (optional)
python -m pip install build twine
python -m build # makes dist/*.whl and *.tar.gz
python -m twine check dist/* # should say PASSED
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****Dependencies:**** the library needs only ****matplotlib**** and ****pandas****. Built with ****setuptools****; tested with ****pytest****. (The animations use `matplotlib`'s`PillowWriter``, i.e. **\*\*Pillow\*\***, which ships with `matplotlib``.)

9. Git / where the code lives

- GitHub repo: ****Prashasti9/Comms_Data_Science****
- Everything is on the `**`main`**` branch (the working branch was merged and deleted). To keep working: edit files, then

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`git add -A && git commit -m "...` && git push origin main`.
- Note: matplotlib PDFs re-render with different bytes each run, so the two
  `.pdf` files may show as "modified" after running the scripts even when the
  content is identical – that's cosmetic; you can `git checkout -- *.pdf` to
  discard that churn.
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10. What's left to do (submission checklist)

- [] Record the ~2-minute video following `**`VIDEO_SCRIPT.md`**`:
install → import → README → `**two favourite charts + aesthetic choices**`
(``share_gap`` and ``growth_line``) → `**two problems + fixes**` (\$7).
- [] Record in `**Zoom → "Record to the Cloud."**`
- [] In the recording's share settings, `**turn OFF the passcode**` (anyone with the link can view).
- [] `**Test the link in an incognito window**` – it must open with no sign-in or password prompt. (This is where marks are most often lost.)
- [] Paste the link into the `**Website URL**` field. `**Due: Fri, Jul 31, 11:59 PM PDT.**`

`**One-line summary for the video:**` `**"simple_viz is a small Python library built on matplotlib and pandas; each of its five functions is the right chart for one question, telling the story that Pinterest's users are global but its revenue is domestic."**`